🌿 Parts of the Plant: Reading

Horticulture 1 · Unit 3 · Level 3 (closer to the class work) · goes with Leaves/Stems/Roots/Flowers worksheets, leaf collection, photosynthesis, Unit 3 quiz

Name: _____

How a Plant Works

Roots anchor a plant in the soil and absorb water and minerals through tiny root hairs. A taproot is one thick main root, like a carrot; fibrous roots are many thin roots, like grass. Some roots store food, and some can be used for propagation. The root cap protects the growing tip as it pushes through soil; it does not absorb water.

Stems hold up leaves and flowers and move materials. Xylem carries water and minerals upward from the roots. Phloem carries food and minerals throughout the plant. In dicot stems, a cambium layer separates the xylem from the phloem in rings; in monocots the vascular bundles are scattered.

Leaves are attached by a petiole, and their main job is photosynthesis. Using light, leaves combine water and carbon dioxide to make sugar (food) and give off oxygen. Gas exchange happens through stomata, tiny openings controlled by guard cells. Through transpiration, leaves give off water vapor. Leaves are described by form (needle, round, ovate), margin (dentate, undulate, incised) and arrangement (opposite, alternate, whorled).

Flowers are the reproductive part. The stamen (male) is made of the anther and filament and produces pollen. The pistil (female) is made of the stigma, style, ovary and ovules. Pollination, often by bees, results in seeds and fruit. Because so many crops depend on bees, declining bee populations are a serious problem for growers.

Check your reading

1. Compare the jobs of xylem and phloem.

2. What are the inputs and products of photosynthesis?

3. Why are bees important to horticulture?